# CEOVIA

## DEVELOPER STARTER PACKAGE — PART 2

Continuation from Section 9 · Prompts, Types, Components, Integrations, Deployment

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**How to Use This Document**

This is Part 2 of the CEOVIA Developer Starter Package. It continues from Section 8 (Daily Build Workflow) in Part 1. Sections, numbering, and formatting are consistent throughout. Use Part 1 and Part 2 together as a single reference during development.

# SECTION 9: EXTENDED PROMPT LIBRARY — PROMPTS 6–15

Prompts 1–5 are in Part 1. The prompts below cover the remaining pages, integrations, and agents. Each prompt is self-contained — paste it directly into Claude or Codex.

## Prompt 6 — Science & Ingredients Page

> **Layer Rule for This Page**
> This is the ONLY page where Layer 2–3 content is permitted. Mechanism-of-action language, study citations, and Evidence Badges all belong here — NOT on the homepage or product pages.

**Copy this prompt:**

```
Build the CEOVIA Science & Ingredients page at src/app/(marketing)/science/page.tsx

Page sections in order:
1. Hero — headline: "The Science Behind CEOVIA" — dark green (#0E5A36) background,
   white text, Cormorant Garamond display font. Subheadline: "190+ bioactive
   compounds. Validated by peer-reviewed research. Extracted at pharmaceutical grade."

2. Origin Story — Sea Buckthorn background: Himalayan origin, 8th century,
   Russian cosmonauts, University of Turku research. Layer 1 language. No claims.

3. The Four-Omega Signature — visual grid showing Omega 3/6/7/9 with
   Layer 2 mechanism description for each. EvidenceBadge component on each.

4. Evidence Library — table with columns: Study | Population | Dose |
   Duration | Key Outcome | Badge. Pull from Sanity using getScienceStudies().
   Each row includes EvidenceBadge + limitation note expandable on click.

5. CO2 Extraction comparison — two-column: CO2 vs Cold Press/Solvent.
   Green column = CEOVIA (checkmarks). Grey column = industry standard (x marks).

6. Bioactive Profile table — Compound | Concentration | Mechanism | Benefit.
   Fetch from Sanity ingredient entity bioactiveProfile field.

7. AEO FAQ Section — structured FAQ accordion. Schema markup: FAQPage JSON-LD.
   Pull questions from Sanity ingredient.aeoFAQBlocks.

8. CTA to Practitioner Portal — "For Healthcare Professionals" section.

Typography: Cormorant Garamond for all section headings. DM Sans for body.
DM Mono for compound names, concentrations, and clinical data.
Background: alternating white (#FFFFFF) and grey (#F4F6F5) sections.
Standard disclaimer block at bottom of page (DisclaimerBlock component).
```

## Prompt 7 — 90-Day System Page

**Copy this prompt:**

```
Build the CEOVIA 90-Day System page at src/app/(marketing)/90-day-system/page.tsx
```

```
Page sections in order:
1. Hero — headline: "The CEOVIA Clinical System — 90 Days to Rejuvenation
   From Within". Full-bleed green background. Primary CTA: "Start the System".

2. Why 90 Days? — explain cellular biology of consistency without mechanism
   language. Layer 1 only. Key message: "True results take consistency."

3. Phase Timeline — three phases with visual timeline component:
   Phase 1 "Reset" (Days 1-30): "Supports your body's natural balance
   and daily function."
   Phase 2 "Restore" (Days 31-60): "Supports skin hydration, energy,
   and immune resilience."
   Phase 3 "Optimize" (Days 61-90): "Supports whole-body balance and
   long-term consistency."
   Visual: horizontal timeline with phase color coding. Phase 1 = green-200,
   Phase 2 = green-400, Phase 3 = green-600.

4. Milestone Journey — four milestone cards at Day 15, 30, 60, 90.
   Each card: day number (large Cormorant Garamond), milestone name,
   what to expect (felt-experience language, NO clinical terms).

5. Daily Ritual — "2 capsules. With breakfast. Every day."
   Keep minimal and elegant. Lifestyle photography context.

6. Progress Markers — what to notice at Day 15/30/60/90.
   Language: "You may notice...", "Many users report..." — never guarantee.

7. Program Selector CTA — embed ProgramSelector component.
   Headline: "Ready to Begin?" Primary: "Start Your 90-Day System".

NO clinical mechanism language on this page. Layer 1 throughout.
No study citations visible to consumer. No percentages or biomarker data.
```

## Prompt 8 — For Clinics / Practitioners Page

> **B2B Priority Rule**
> This page must be accessible from top navigation as an explicit link. It is a primary revenue channel, not a secondary page. Build it to the same quality standard as the homepage.

**Copy this prompt:**

```
Build the CEOVIA For Clinics page at src/app/(marketing)/for-clinics/page.tsx

Page sections in order:
1. Hero — headline: "Clinical-Grade Nutrition for Your Practice"
   Subheadline: "CEOVIA supports practitioners across dermatology,
   gynecology, cardiology, and integrative medicine."
   Primary CTA: "Request Professional Access"
   Secondary CTA: "Download Clinical Summary" (triggers PDF download)
   Background: white with green (#0E5A36) accent elements

2. Why Practitioners Choose CEOVIA — four specialty use cases:
   Dermatology | Gynecology / Menopause | Cardiology | Integrative Medicine
   Each: specialty name (h3), patient profile, clinical relevance (Layer 2 language).

3. Clinical Evidence Summary — three key studies in card format:
   Chan et al. (2024) — skin hydration RCT
```

```
   Danaei et al. (2017) — cardiovascular markers
   Larmo et al. (2010) — mucosal health
   Each card: citation, finding (max 200 chars), Gold EvidenceBadge,
   limitation note. Link to full Science page.

4. What Practitioners Get — feature list:
   Clinical dossier PDF | Patient education kits | Wholesale pricing
   | Practitioner portal access | CoA per batch

5. 3-Step Access Process — Step 1: Submit credentials | Step 2: 48hr
   verification | Step 3: Full access + account manager assigned

6. Registration Form — fields: Name, Practice Name, Specialty,
   Country, Email, Phone, Volume Estimate (dropdown).
   On submit: POST to /api/crm/practitioner-registration
   Create HubSpot contact with lifecycle stage "Lead", list "B2B Practitioners"

7. Trust signals — WHO cGMP, Halal cert, Third-party tested, FDA GRAS
```

## Prompt 9 — For Distributors Page

**Copy this prompt:**

```
Build the CEOVIA For Distributors page at src/app/(marketing)/for-distributors/page.tsx

Page sections in order:
1. Hero — headline: "Bring CEOVIA to Your Market"
   Subheadline: "Global distribution opportunities across GCC, Europe,
   Asia-Pacific, and the Americas."
   Primary CTA: "Apply for Partnership"
   Secondary CTA: "Download Volume Pricing Guide"

2. Partnership Types — four cards:
   Retail Distribution | Clinical/Pharmacy Channel
   | International Master Distributor | Exclusive Territory
   Each: description, minimum order indication, territory availability

3. Why Partner with CEOVIA — three columns:
   Premium Positioning: defensible differentiation, clinical credibility
   Commercial Support: marketing assets, compliance-cleared content
   Growth Market: premium wellness demand, multi-market architecture

4. Market Availability Map — static visual showing open vs. allocated
   territories. Label: GCC (allocated), USA (allocated), EU (open),
   UK (open), APAC (open), LatAm (open)

5. Partnership Application Form — Company Name, Market/Region,
   Channel Type, Estimated Annual Volume, Contact Name, Email, Phone.
   On submit: POST to /api/crm/distributor-application
   HubSpot: lifecycle stage "Lead", list "B2B Distributors",
   deal stage "New Partnership Inquiry"

6. Footer note: "We respond to all partnership inquiries within 48 hours."
```

## Prompt 10 — FAQ Page

**Copy this prompt:**

```
Build the CEOVIA FAQ page at src/app/(marketing)/faq/page.tsx

Structure: accordion FAQ grouped by category.

Category 1: "About the Product"
  Q: What is CEOVIA?
  Q: What makes Sea Buckthorn seed oil different from other supplements?
  Q: How many bioactive compounds does CEOVIA contain?
  Q: Is CEOVIA Halal certified?
  Q: Where is CEOVIA manufactured?

Category 2: "How to Use"
  Q: When is the best time to take CEOVIA and should I take it with food?
  Q: How long does it take to see results?
  Q: Can I take this with other supplements?
  Q: Is CEOVIA suitable for all ages?
  Q: Can I take more than 2 capsules per day?

Category 3: "Safety & Interactions"
  Q: Is CEOVIA safe for daily use?
  Q: Are there any contraindications?
  Q: Is CEOVIA safe during pregnancy?
  Q: I am on anticoagulants — can I take CEOVIA?

Category 4: "Orders & Subscription"
  Q: How does the subscription work?
  Q: Can I pause or cancel my subscription?
  Q: What is your returns policy?
  Q: How long does delivery take?

All answers: Layer 1 language only. No disease names.
Safety answers: always include "consult your healthcare provider" qualifier.
Schema markup: FAQPage JSON-LD for all questions.
Standard disclaimer block at bottom of page.
```

## Prompt 11 — Compliance Critic Scoring Endpoint

**Copy this prompt:**

```
Build a Next.js 15 API route at src/app/api/ai/compliance/route.ts

This is the Compliance Critic Agent scoring endpoint.
Called by: content creation tools, SEO Content Agent, WhatsApp agent
review, and the manual compliance check in Sanity Studio.

POST /api/ai/compliance
Body: { content: string, layer: "layer1"|"layer2"|"layer3", market: string }

Scoring logic:
Start score: 70

PROHIBITED terms (any found → score = Math.min(score, 79)):
  Medical verbs: treats|cures|heals|prevents|reverses|eliminates
  Clinical descriptors: immunomodulatory|neuroprotective|anti.tumor
  Condition names: arthritis|hepatitis|dementia|melasma|ulcer|eczema|
    psoriasis|acne (as treatment)|rosacea|schizophrenia
  Pathogens: herpes|influenza|e\.coli|adenovirus|diphtheria|staph
  Drug language: arrests|halts|remission|reversal
  Layer 1 violations: collagen.synthesis|hormonal.balance|NF-κB|
```

```
    cardiovascular.markers|neuroprotect

YELLOW FLAG terms (score -= 5 per occurrence, only if score >= 80):
  boosts?(?= immune| energy)|stimulates|controls|improves(?! overall)
  \d+%(?! of)|proven to|clinically proven|guaranteed

POSITIVE signals (score += 5 each, cap at 100):
  \bsupports\b | \bhelps maintain\b | \bdesigned to support\b
  \bformulated to support\b | \bcrafted to support\b
  Disclaimer present: "not intended to diagnose" → +10
  No disease names → +10

Auto-fix replacements (apply when score 80-94):
  "treats" → "supports"
  "cures" → "is designed to support"
  "prevents" → "helps maintain"
  "boosts" → "supports"
  "stimulates" → "supports"
  "improves" → "supports"
  "immunomodulatory" → "supports your body's natural resilience"
  "neuroprotective" → "supports healthy brain and nervous system function"

Return: { score, status: "pass"|"fixed"|"flagged",
  violations: string[], suggestions: string[],
  autoFixedContent?: string }

Log all requests to PostgreSQL compliance_log table:
  (id, content_hash, score, status, violations[], market, layer,
   created_at, agent_id)
```

## Prompt 12 — WhatsApp Sales Agent API

**Phase 1 Agent — Critical**

This is one of the four Phase 1 agents. It must be built and tested before launch. The WhatsApp Sales Agent is a SALES DESK, not a general assistant. Scope is strictly: qualify, educate, handle objections, convert, hand off. Out of scope: medical advice, drug interactions, regulatory statements.

**Copy this prompt:**

```
Build a Next.js API route at src/app/api/whatsapp/webhook/route.ts
This handles incoming WhatsApp Business API messages.

Step 1: Webhook verification (GET handler)
  Verify the hub.challenge from Meta webhook setup
  Check hub.verify_token matches WHATSAPP_WEBHOOK_SECRET env var

Step 2: Message handler (POST handler)
  Parse incoming message from Meta webhook payload
  Extract: from (phone), message.text.body, timestamp
  Retrieve conversation history from Redis (key: "wa:{phone}")

Step 3: Build agent context
  System prompt:
  "You are the CEOVIA WhatsApp Sales Agent — a sales desk, not a general
   assistant. You qualify leads, explain the 90-day system, handle the six
   standard objections (price, efficacy, safety, competitors, past failures,
   FDA), and guide users to purchase. You NEVER provide medical advice,
```

```
   diagnose, or recommend for specific conditions. You NEVER discuss drug
   interactions. Layer 1 language only: supports, helps maintain, designed
   to support. Always end with: Results may vary. Consult a healthcare
   professional if needed.

   Product: CEOVIA 90-Day System (AED 825 / $225) - 180 softgel capsules
   of supercritical CO2 extracted Sea Buckthorn seed oil. Take 2 capsules
   daily with a meal containing healthy fats.

   Conversation phases:
   Phase 1 (Greeting): Ask if exploring for personal wellness, practitioner,
   distributor, or investor.
   Phase 2 (Needs Assessment): Primary goal? Previous supplement experience?
   Phase 3 (Education & Objection Handling): Use objection tree below.
   Phase 4 (Product Selection): Recommend 30/60/90-day or subscription.
   Phase 5 (Conversion): Provide Shopify checkout link for selected variant.

   ESCALATE IMMEDIATELY (set escalation flag, notify human) when:
   - User mentions a specific medication by name
   - User asks about medical diagnosis or treatment
   - Negative sentiment detected (scam, doesn't work, complaint)
   - B2B inquiry (clinic, distributor, bulk order)
   - User expresses pregnancy or nursing
   - Hot lead score >85 (buy signals in 3+ messages)"

Step 4: Call OpenAI GPT-4o-mini (temperature 0.3, max_tokens 300)

Step 5: Send response via WhatsApp Business API
  POST https://graph.facebook.com/v18.0/{WHATSAPP_PHONE_NUMBER_ID}/messages

Step 6: Update conversation in Redis (TTL: 90 days)
  Key: "wa:{phone}", Value: JSON array of {role, content, timestamp}

Step 7: Log interaction to PostgreSQL whatsapp_conversations table
  Create/update HubSpot contact via HubSpot Contacts API
  Set properties: whatsapp_phone, lead_source: "whatsapp",
  lifecycle_stage based on conversation phase
```

## Prompt 13 — HubSpot CRM Integration

**Copy this prompt:**

```
Create src/lib/hubspot/client.ts
HubSpot integration for CEOVIA.

Install: npm install @hubspot/api-client

Functions to implement:

1. createOrUpdateContact(data: {
     email: string, firstName?: string, lastName?: string,
     phone?: string, market?: string, userType?: "consumer"|"clinic"|"distributor",
     leadSource?: string, lifecycleStage?: string
   })
   → Upsert contact. Properties to set:
   ceovia_user_type, ceovia_market, ceovia_lead_source,
   ceovia_first_contact_date, ceovia_qualification_phase

2. updateLeadScore(contactId: string, score: number, signals: string[])
```

```
   → Update ceovia_lead_score and ceovia_lead_signals properties

3. logConversation(contactId: string, channel: "whatsapp"|"chat",
   summary: string, phase: string)
   → Create HubSpot engagement (NOTE type) on contact

4. triggerWorkflow(contactId: string, workflowId: string)
   → Trigger HubSpot enrollment workflow

5. createDeal(contactId: string, data: {
   dealName: string, amount: number, currency: string,
   stage: string, source: string
   })
   → Create deal and associate with contact

Error handling: all functions must catch and log errors without
crashing the calling agent. HubSpot failures are non-blocking.
```

## Prompt 14 — Geo-Detection & Multi-Market Logic

**Copy this prompt:**

```
Create src/lib/utils/geo.ts

Implement geo-detection for CEOVIA multi-market pricing and compliance.

Types:
type Market = "uae" | "usa" | "uk" | "eu" | "sa" | "global"
type Currency = "AED" | "USD" | "GBP" | "EUR" | "SAR"

type MarketConfig = {
  market: Market
  currency: Currency
  currencySymbol: string
  locale: string
  discountText: string  // e.g. "Save 20%"
  trustSignals: string[] // market-specific certs to show
  whatsappEnabled: boolean
  arabicSupport: boolean
}

function detectMarket(request: Request): Market
  → Read Cloudflare cf-ipcountry header (X-Vercel-IP-Country fallback)
  → Map country codes: AE/SA/QA/KW/BH/OM → uae/sa/global
  → US → usa, GB → uk, EU country codes → eu
  → Default: "global" (shows USD)

function getMarketConfig(market: Market): MarketConfig
  → Return full config object per market

Market configs:
UAE: AED, Halal cert prominent, WhatsApp primary, Arabic support
USA: USD, FDA disclaimer prominent, subscription emphasis
UK: GBP, MHRA note, editorial positioning
EU: EUR, EFSA disclaimer, single-purchase preference
SA: SAR, SFDA note, Halal cert, Arabic required
Global: USD, generic trust signals

function formatPrice(amount: number, config: MarketConfig): string
```

```
  → Format with Intl.NumberFormat using config.locale and config.currency

Export a getMarket() server function that reads from request headers.
Export a useMarket() client hook that reads from a cookie set at edge.
The cookie: "ceovia_market" — set in middleware.ts on first visit.
```

## Prompt 15 — Sanity GROQ Queries

**Copy this prompt:**

```
Create src/lib/sanity/queries.ts
All GROQ queries for the CEOVIA Sanity CMS.

Install: npm install next-sanity @sanity/client

Queries to implement (each as a named export):

1. HOMEPAGE_QUERY — fetch hero content, benefit bullets, 90-day phases,
   trust signals from the "homepage" singleton document

2. PRODUCT_QUERY — fetch product by slug with all variants, layer1Benefits,
   usageProtocol, disclaimer, and referenced ingredient names/slugs

3. INGREDIENT_QUERY — fetch full ingredient including bioactiveProfile,
   clinicalStudies (filtered: studyType and evidenceBadge),
   layer1Claims, aeoFAQBlocks. Filter by slug.

4. SCIENCE_PAGE_QUERY — fetch all ingredients + their clinicalStudies
   where evidenceBadge in ["gold","silver"] only. Include compliantFraming.

5. APPROVED_CLAIMS_QUERY(market: string) — fetch claims where
   status == "approved" and market in applicableMarkets[]
   Returns: id, text, category, layer, evidenceBadge
   Used by: AI middleware to ground all responses

6. BLOG_POSTS_QUERY — fetch published posts where
   complianceStatus == "pass" || "fixed"
   Order by publishedAt desc. Include: title, slug, excerpt,
   featuredImage, occasionTags, layerClassification

7. BLOG_POST_QUERY — fetch single post by slug with full body,
   evidenceBlocks, faqBlocks, relatedIngredients[]->name

8. MARKET_RULE_QUERY(market: string) — fetch single marketRule
   by marketCode. Returns disclaimer, prohibitedTerms[], languages[]

All queries: use groq tag template. Export types alongside each query.
Create getCachedClient() with revalidate: 3600 (1hr) for static queries.
Use revalidate: 60 for blog posts and product queries.
```

# SECTION 10: TYPESCRIPT TYPES REFERENCE

Create src/types/ceovia.ts with these types. They are used across all components, API routes, and Sanity queries. Paste these into Codex with: "Add these TypeScript types to src/types/ceovia.ts. Do not modify the type names — they are referenced throughout the codebase."

## 10.1 Market & Currency Types

```
export type Market = 'uae' | 'usa' | 'uk' | 'eu' | 'sa' | 'global'
export type Currency = 'AED' | 'USD' | 'GBP' | 'EUR' | 'SAR'
export type ContentLayer = 'layer1' | 'layer2' | 'layer3'
export type EvidenceBadge = 'gold' | 'silver' | 'bronze' | 'hypothesis'
export type ComplianceStatus = 'pass' | 'fixed' | 'flagged' | 'pending'
export type UserType = 'consumer' | 'clinic' | 'distributor' | 'investor'
```

## 10.2 Product Types

```
export interface ProductVariant {
  id: string
  label: string
  shopifyVariantId: string
  capsules: number
  days: number
  priceAED: number
  priceUSD: number
  priceGBP: number
  priceEUR: number
  perDayCostDisplay: string
  isRecommended?: boolean
  isSub?: boolean
  tagline: string
  subscriptionSave?: string
}
```

```
export interface Product {
  _id: string
  name: string
  slug: string
  shopifyId: string
  variants: ProductVariant[]
  layer1Benefits: Array<{ benefit: string; studyRef?: string }>
  phases: Array<{ phase: string; days: string; consumerCopy: string; checkpointNote?:
string }>
  usageProtocol: {
    capsulesPerDay: number
    timing: string
    minimumDays: number
    maxSafeDose: string
  }
  disclaimer: string
  markets: Market[]
}
```

## 10.3 Compliance Types

```
export interface ComplianceResult {
  score: number
  status: ComplianceStatus
  violations: string[]
  suggestions: string[]
  autoFixedContent?: string
}
```

```
export interface ApprovedClaim {
  id: string
  text: string
  category: string
  layer: ContentLayer
  evidenceBadge: EvidenceBadge
  applicableMarkets: Market[]
  sourceStudy?: string
}
```

## 10.4 AI Agent Types

```
export interface AIGovernanceContext {
  userMarket: Market
  userType: UserType
  conversationHistory: Array<{ role: "user" | "assistant"; content: string; timestamp:
string }>
  approvedClaims: ApprovedClaim[]
  productData: Product[]
  sessionId: string
  escalationFlag?: boolean
  escalationReason?: string
}
```

```
export interface ChatMessage {
  id: string
  role: "user" | "assistant" | "system"
  content: string
  timestamp: string
  complianceScore?: number
  escalationFlag?: boolean
}
```

# SECTION 11: COMPONENT SPECIFICATIONS

These specifications define the exact behaviour and props for each reusable component. Give these to Codex alongside the prompt: "Build this component exactly as specified. Use the CEOVIA Tailwind config for all styling. TypeScript required."

## 11.1 Core UI Components

| Component | Props | Behaviour & Notes |
|---|---|---|
| **Button** | variant: "primary"\|"secondary"\|"tertiary" size: "sm"\|"md"\|"lg" loading?: boolean fullWidth?: boolean | primary = green (#0E5A36). secondary = outlined green. tertiary = orange (#F4A21E). Loading state: spinner replaces label. fullWidth: 100% width on mobile. Pill border-radius always. |
| **EvidenceBadge** | type: EvidenceBadge citation?: string onClick?: () => void | Gold/Silver/Bronze/Hypothesis with correct colours from design tokens. Click shows tooltip with citation, finding (max 200 chars), limitation note. Pulse animation when first rendered on Science page. |
| **DisclaimerBlock** | market?: Market variant: "full"\|"short" | Required on all product pages and blog posts. Full: complete FDA/EFSA/MOH text. Short: one sentence. Market-specific text loaded from marketRule Sanity query. Never optional. |
| **LayerTag** | layer: ContentLayer | Visual pill showing content layer classification. Layer 1 = green, Layer 2 = amber, Layer 3 = blue. Used in Sanity Studio and Practitioner Portal to signal which audience can see this content. |
| **Accordion** | items: Array<{question, answer}> schema?: boolean | Smooth CSS height transition. Single open at a time (FAQ mode) or multiple (Science mode). schema=true injects FAQPage JSON-LD. Used on FAQ page and Science page. |
| **TrustBadge** | type: "halal"\|"gmp"\|"co2"\|"nongmo"\|"tested" size: "sm"\|"md" | Certification badge with icon and label. Renders from a static map of type → icon + label text. Used in TrustBar and product page trust grid. |
| **PriceDisplay** | amount: number market: Market perDay?: string sub?: boolean | Formats price using Intl.NumberFormat with market locale. Shows per-day cost below price. sub=true shows save% badge. Reads market config from geo context. |

## 11.2 Section Components

| Component | Key Props | Build Notes |
|---|---|---|

| | | |
|---|---|---|
| **Hero** | headline, subheadline primaryCTA, secondaryCTA heroImage, theme: "dark"\|"light" | theme=dark: green bg + white text. theme=light: white bg + green text. Framer Motion: headline fadeUp 0.6s, subheadline 0.8s, CTAs 1.0s, trust pills 1.2s. Hero image right-aligned on desktop, hidden below fold on mobile. |
| **NinetyDaySystem** | phases: Phase[] showCTA?: boolean | Horizontal timeline on desktop, vertical accordion on mobile. Phase colours: Reset=green-200, Restore=green-400, Optimize=green-600. Checkpoint note italic below each phase. NO clinical language — passes layer1 check before render. |
| **ProgramSelector** | variants: ProductVariant[] defaultVariant?: string market: Market | Pre-selected: 90-day. On select: update price display, tagline, CTA text, per-day cost. Framer Motion scale(1.02) on selected card. 90-day card: gold border + "Recommended" badge. Sub card: green save% badge. CTA fires addToCart(). |
| **TrustBar** | market?: Market | Horizontal infinite scroll ticker. Green bg, white text. 8 items × 2 for seamless loop. Pause on hover. market prop controls which Halal cert note to show. Aria-label set. |
| **ScienceCTA** | variant: "consumer"\|"professional" | consumer: "Backed by Research" — links to /science. professional: "For Professionals" — links to /for-clinics. Two separate visual treatments. Always positioned as penultimate page section. |
| **ChatWidget** | market: Market defaultOpen?: boolean | Fixed bottom-right. Collapsed: circular green icon. Expanded: chat drawer 380px wide, full height. Calls /api/ai/chat. Streaming response. Shows escalation message if flag triggered. WhatsApp button in footer of widget for handoff. |

# SECTION 12: AI GOVERNANCE MIDDLEWARE

The AI middleware is the governance layer between user requests and OpenAI. Every AI response must pass through it. Nothing bypasses this. Copy this prompt to build the middleware file.

**Copy this prompt:**

```
Create src/lib/ai/middleware.ts
This is the CEOVIA AI Governance Context builder.
Called by: Smart Chat Agent, WhatsApp Sales Agent, Compliance Critic.

Function: buildGovernanceContext(input: {
  userMessage: string
  market: Market
  userType: UserType
  sessionId: string
  conversationHistory: ChatMessage[]
}): Promise<AIGovernanceContext>

Steps:
1. Extract implied claims from userMessage (keyword matching)
2. Fetch approvedClaims from Sanity for this market (APPROVED_CLAIMS_QUERY)
3. Check all implied claims exist in approvedClaims
4. If any missing: set escalationFlag=true, escalationReason="unapproved_claim"
5. Fetch product data from Sanity (PRODUCT_QUERY for active product)
6. Build and return AIGovernanceContext object

Hard blocks (return safe fallback response, do not call OpenAI):
  - User mentions: diagnose|treat|cure|prescription|dose of [drug name]
  - User mentions specific medications: warfarin|metformin|statin|ssri|etc
  - User expresses: pregnant|nursing|breastfeeding (→ consult HCP message)

Safe fallback response:
"For specific health concerns or medication interactions, I recommend
speaking with your healthcare provider. I can connect you with a CEOVIA
specialist for a personal consultation — would that be helpful?"

Compliance post-validation (after OpenAI response):
  Call /api/ai/compliance with the generated response
  If score < 80: replace with safe fallback, log violation
  If score 80-94: use autoFixedContent, log the fix
  If score >= 95: use response as-is
```

# SECTION 13: POSTGRESQL DATABASE SCHEMA

Copy this prompt to Codex to create the initial database migration:

**Copy this prompt:**

```
Create database/migrations/001_initial_schema.sql
PostgreSQL 16 schema for CEOVIA governance database.

Tables:

1. compliance_log
   id UUID PRIMARY KEY DEFAULT gen_random_uuid()
   content_hash VARCHAR(64) NOT NULL
   score INTEGER NOT NULL CHECK (score BETWEEN 0 AND 100)
   status VARCHAR(20) NOT NULL CHECK (status IN ("pass","fixed","flagged","pending"))
   violations TEXT[] DEFAULT '{}'
   market VARCHAR(10) NOT NULL
   layer VARCHAR(10) NOT NULL
   agent_id VARCHAR(50)
   created_at TIMESTAMPTZ DEFAULT NOW()
   -- 7-year retention required for regulatory audit

2. ai_interactions
   id UUID PRIMARY KEY DEFAULT gen_random_uuid()
   session_id VARCHAR(100) NOT NULL
   channel VARCHAR(20) NOT NULL CHECK (channel IN ("chat","whatsapp","api"))
   user_message_hash VARCHAR(64) NOT NULL
   response_hash VARCHAR(64) NOT NULL
   claims_used TEXT[] DEFAULT '{}'
   compliance_score INTEGER
   escalation_flag BOOLEAN DEFAULT FALSE
   escalation_reason TEXT
   market VARCHAR(10)
   user_type VARCHAR(20)
   created_at TIMESTAMPTZ DEFAULT NOW()
   -- Store hashes only, never raw message content

3. whatsapp_conversations
   id UUID PRIMARY KEY DEFAULT gen_random_uuid()
   phone_hash VARCHAR(64) NOT NULL
   hubspot_contact_id VARCHAR(50)
   conversation_phase VARCHAR(30)
   lead_score INTEGER DEFAULT 0
   escalation_flag BOOLEAN DEFAULT FALSE
   last_message_at TIMESTAMPTZ
   created_at TIMESTAMPTZ DEFAULT NOW()
   updated_at TIMESTAMPTZ DEFAULT NOW()

4. content_approvals
   id UUID PRIMARY KEY DEFAULT gen_random_uuid()
   sanity_document_id VARCHAR(100) NOT NULL
   document_type VARCHAR(50) NOT NULL
   compliance_score INTEGER NOT NULL
   approved_by VARCHAR(100)
   approved_at TIMESTAMPTZ
   notes TEXT
   created_at TIMESTAMPTZ DEFAULT NOW()
```

```
Indexes: create on session_id, phone_hash, sanity_document_id,
created_at DESC (all tables), market (compliance_log)
RLS: enable row-level security on all tables
```

# SECTION 14: TESTING PROTOCOL

Run this testing sequence before every deployment. Use the prompts below to ask Claude/Codex to write the tests for you.

## 14.1 Compliance Tests — Highest Priority

> **Why This Test Suite Is Non-Negotiable**
> A compliance violation reaching a live page is a regulatory event. Run these tests before every deployment, no exceptions. If any compliance test fails, the deployment is blocked.

**Copy this prompt to generate the compliance test suite:**

```
Write a Jest test suite for the CEOVIA Compliance Critic at
src/app/api/ai/compliance/route.ts

Tests to include:

HIGH RISK — all must score < 80 and status = "flagged":
  "CEOVIA treats gastric ulcers"
  "Effective against influenza and herpes"
  "Exhibits neuroprotective and anti-tumor activity"
  "Prevents premature photo-aging"
  "Reduces hyperpigmentation and Melasma"
  "Reverses senile dementia"
  "Cures hepatitis"

YELLOW FLAG — must score 80-94 and status = "fixed":
  "CEOVIA boosts your immune system"
  "Improves skin hydration"
  "Stimulates collagen production"

COMPLIANT — must score >= 95 and status = "pass":
  "Designed to support skin hydration and elasticity"
  "Helps maintain healthy energy levels"
  "Formulated to support your body's natural resilience"
  "Supports joint comfort and flexibility"

AUTO-FIX accuracy — verify autoFixedContent replaces:
  "boosts" → "supports"
  "treats" → "supports"
  "prevents" → "helps maintain"
```

## 14.2 Component Tests

| Component | What to Test |
|---|---|
| **ProgramSelector** | Default selection = 90-day. Selecting 30-day updates price, tagline, and CTA text correctly. Gold border only on 90-day card. Per-day cost calculates correctly for each variant. Market prop changes displayed currency. |

| | |
|---|---|
| **EvidenceBadge** | Gold badge = #D4A857. Silver = #87A878. Bronze = #C4956A. Hypothesis = #9DAAA2. Citation tooltip appears on click. Tooltip disappears on outside click. |
| **DisclaimerBlock** | Renders for all markets. Never returns null. Full variant includes "not intended to diagnose, treat, cure, or prevent any disease." Short variant is under 100 characters. |
| **ChatWidget** | Collapsed state shows green icon. Expanded state shows chat drawer. Input submission calls /api/ai/chat. Escalation message renders when escalationFlag = true. WhatsApp handoff button is visible. |
| **Navbar** | "For Clinics" and "For Distributors" links are always visible in nav — not inside dropdowns. Primary CTA button is always visible. Mobile: hamburger opens drawer with all links. |

## 14.3 Pre-Deploy Manual Checklist

**Run through this checklist manually before every production deploy:**

| | Check Item |
|---|---|
| ☐ | Homepage: no disease names visible anywhere above or below fold |
| ☐ | Homepage: "collagen synthesis", "hormonal balance", "cardiovascular markers" do NOT appear |
| ☐ | Product page: disclaimer block is visible above the footer on all variants |
| ☐ | Product page: 90-day variant is pre-selected on load |
| ☐ | Navigation: "For Clinics" and "For Distributors" are visible text links in top nav |
| ☐ | Science page: every study shows citation + finding + limitation note |
| ☐ | Science page: Evidence Badges match study type (Gold = RCT, Silver = in-vitro only) |
| ☐ | Chat widget: medical question triggers escalation message, not a medical answer |
| ☐ | Mobile (390px): all CTAs in thumb zone (bottom 40% of viewport) |
| ☐ | Mobile: Lighthouse score ≥ 90 (run PageSpeed Insights on homepage URL) |
| ☐ | UAE visitor: AED pricing displayed, Halal cert badge visible, WhatsApp CTA present |
| ☐ | US visitor: USD pricing, FDA disclaimer visible |
| ☐ | No "anti-tumor", "treats hepatitis", "prevents dementia" in any page source |
| ☐ | Dr. Oz reference removed from all pages and documents |
| ☐ | Consent banner appears for EU visitors before any tracking activates |

# SECTION 15: DEPLOYMENT CHECKLIST

Complete all items in this section before going live. Items marked URGENT must be completed before launch. Items marked PRE-LAUNCH must be done in the 48 hours before go-live.

## 15.1 Infrastructure Setup

| Priority | Service | Action Required |
|---|---|---|
| URGENT | Vercel | Connect GitHub repo. Set environment variables from Section 6. Enable Edge Network. Set NEXT_PUBLIC_SANITY_PROJECT_ID. |
| URGENT | Sanity | Create project at sanity.io. Import all five schemas. Create "homepage" singleton document. Populate product document with CEOVIA 90-day system data. |
| URGENT | Shopify Plus | Create store at ceovia.myshopify.com. Add products with correct variant IDs. Enable Storefront API. Configure Shopify B2B for clinic channel. |
| URGENT | Cloudflare | Point domain to Vercel. Enable WAF. Configure DLP worker for /api/ai/* routes. Set geographic routing rules for GCC, EU, US. |
| PRE-LAUNCH | WhatsApp | Submit WhatsApp Business API application via Meta. Requires: business verification, phone number, messaging template approval. Allow 5–7 business days. |
| PRE-LAUNCH | HubSpot | Create custom properties: ceovia_user_type, ceovia_market, ceovia_lead_score, ceovia_qualification_phase. Create lists: B2B Practitioners, B2B Distributors, Consumer 90-Day. |
| PRE-LAUNCH | Supabase | Create project. Run migration 001_initial_schema.sql. Enable Row Level Security. Obtain DATABASE_URL connection string. |
| PRE-LAUNCH | Algolia | Create index: ceovia_products. Configure searchable attributes: name, layer1Benefits.benefit, ingredients.name. Set ranking: relevance + variants.days desc. |
| PRE-LAUNCH | Upstash Redis | Create database. Set UPSTASH_REDIS_REST_URL and TOKEN in Vercel env vars. Test connection with a ping. |

## 15.2 Content Pre-Load Checklist

**Load this content into Sanity Studio before go-live:**

- Product document: CEOVIA 90-Day System with all 4 variants, prices in AED/USD/GBP/EUR, 90-day phases, usage protocol, and disclaimer
- Ingredient document: Sea Buckthorn Seed Oil — all bioactive profile entries, minimum 5 clinical studies with evidence badges and limitation notes
- Approved claims: minimum 15 Layer 1 consumer claims approved for "global" market
- Market rules: create rules for USA, EU, UAE, UK, Saudi Arabia — each with correct disclaimer and prohibited terms
- Homepage singleton: hero content, 4 benefit bullets, 90-day phase summaries
- FAQ entries: minimum 15 questions across 4 categories, all Layer 1 compliant

## 15.3 DNS & Domain Configuration

| Domain | Market | Configuration |
|---|---|---|
| **ceovia.com** | US / Global | Primary. Point CNAME to Vercel deployment. USD default. |
| ceovia.ae (Phase 2) | UAE / GCC | AED pricing, Arabic support, Halal cert prominent. WhatsApp-first. |
| ceovia.co.uk (Phase 2) | United Kingdom | GBP pricing, MHRA disclaimer, editorial positioning. |
| ceovia.eu (Phase 2) | EU (GDPR) | EUR pricing, EFSA claims only, GDPR consent banner mandatory. |
| pro.ceovia.com | Global B2B | Gated practitioner portal. Registration wall on all content. |

## 15.4 Launch Day Sequence

**Execute in this exact order on launch day:**

1. Run full test suite: npm run test — all tests must pass
2. Run compliance test suite: npm run test:compliance — zero failures
3. Run Pre-Deploy Manual Checklist (Section 14.3) — all 16 items checked
4. Final Lighthouse run: score ≥ 90 on mobile for homepage
5. Set Vercel deployment to production branch
6. Verify environment variables are set correctly in Vercel dashboard
7. Deploy: git push origin main → triggers Vercel production build
8. DNS propagation: verify ceovia.com resolves to Vercel deployment
9. Smoke test live site: homepage, product page, FAQ, For Clinics, chat widget
10. Verify geo-detection: test UAE IP (AED pricing) and US IP (USD pricing)
11. Submit XML sitemap to Google Search Console
12. Monitor Vercel logs for first 2 hours — watch for API errors

# SECTION 16: PHASE 1 AGENT READINESS TRACKER

**Phase 1 Scope Reminder**

Only four agents are in Phase 1: Smart Chat Agent, WhatsApp Sales Agent, Compliance Critic Agent, SEO Content Agent. Everything else is Phase 2 or 3. Do not build Phase 2 agents until all Phase 1 agents are live, tested, and stable for 14+ days.

| Agent | Build Status | Live Status | Definition of Done |
|---|---|---|---|
| **Smart Chat Agent (Website)** | □ Not started □ In progress □ Complete | □ Staging □ Production | Answers product questions. Returns Layer 1 language only. Escalation flag triggers on medical questions. Logs to PostgreSQL. |
| **WhatsApp Sales Agent** | □ Not started □ In progress □ Complete | □ Staging □ Production | Handles 5 conversation phases. Objection tree functional. Human handoff triggers correctly. Response time < 2 minutes. HubSpot contact created on first message. |
| **Compliance Critic Agent** | □ Not started □ In progress □ Complete | □ Staging □ Production | Scores content 0–100. All 7 HIGH RISK test cases return "flagged". All 3 YELLOW FLAG cases return "fixed". All 3 COMPLIANT cases return "pass". Auto-fix accuracy > 95%. |
| **SEO Content Agent (Prompt → Draft)** | □ Not started □ In progress □ Complete | □ Staging □ Production | Generates blog drafts using SEO prompt template. Submits to Compliance Critic before Sanity. First-pass approval rate > 80%. Occasion tags included. Disclaimer auto-appended. |

# SECTION 17: QUICK REFERENCE CARD

Keep this section open in a separate tab during development. It covers the decisions you will need to make most frequently.

## 17.1 Standard Component Prompt Opening

**Use this opening for every component prompt to Claude/Codex**

"Build the [ComponentName] component for CEOVIA following the Tailwind design system (tailwind.config.v2.js). Primary green: #0E5A36. Gold: #D4A857. Orange: #F4A21E. Page background: white #FFFFFF. Section background: grey #F4F6F5. Font-display (Cormorant Garamond) for headings. Font-sans (DM Sans) for body. Layer 1 language only unless this is the Science or Practitioner page — no disease names, no mechanism claims."

## 17.2 Approved Verb Reference

| ✓ USE — Approved Verbs (Layer 1) | ✕ NEVER USE — Blocked Terms |
|---|---|
| "Supports" / "Designed to support" | "Treats" / "Cures" / "Heals" |
| "Helps maintain" | "Prevents" / "Reverses" / "Eliminates" |
| "Formulated to support" | "Immunomodulatory" / "Neuroprotective" |
| "Crafted to support" | "Anti-tumor" — blocked on ALL layers |
| "Traditionally studied for" | Disease names (arthritis, hepatitis, dementia, melasma) |
| "Peer-reviewed research suggests" | Pathogens (herpes, influenza, E. coli, diphtheria) |
| "Results may vary" | "Boosts" as primary claim verb |

## 17.3 Brand Colour Quick Reference

| Token | Hex | Use For |
|---|---|---|
| **green-600** | #0E5A36 | Headings, primary CTAs, nav, section dividers, dark hero bg. 20% of layout. |
| **gold.400** | #D4A857 | Evidence badges (Gold), "Recommended" card border, heading underlines. Sparingly. |
| **orange.500** | #F4A21E | Secondary CTAs, icons, key data highlights. Max 1 per viewport. Never as bg. |
| **white** | #FFFFFF | Page background. 70% of all layouts. Clinical, modern, premium. |
| **grey.50** | #F4F6F5 | Section backgrounds, cards, UI blocks. 7% of layouts. |
| **grey.200** | #C8D1CB | Borders, input outlines, dividers, table row separators. |
| **text.primary** | #2B2B2B | All body copy. Deep charcoal, not pure black. |

## 17.4 Page → Content Layer Map (One-Line Version)

| Page | Layer Rule |
|---|---|
| **Homepage** | Layer 1 ONLY — benefit language, no clinical terms, no condition names |
| **Product Pages** | Layer 1 ONLY — same as homepage, science links to /science |
| **Blog / Journal** | Layer 1–2 — educational depth ok, Evidence Badges required, disclaimer mandatory |
| **Science Page** | Layer 2–3 — mechanism language, citations, full context, limitation notes |
| **Practitioner Portal** | Layer 2–3 — gated, full clinical language, contraindications included |
| **Social / Community** | Layer 1 ONLY — all UGC must have approvedForSite=true in Sanity |
| **FAQ** | Layer 1 — no disease names, "consult your healthcare provider" on safety Qs |

— End of Developer Starter Package Part 2 —

Continue to: CEOVIA_AI_Agent_Ecosystem_V3.docx and CEOVIA_Technical_Platform_Blueprint_V3.docx